Discrete mathematics for 26-DS(Eve)

Assignment: Sets, Relations, and Functions

Instructions

- Attempt **all** questions.
- Show all steps clearly; justify each answer.
- Work out your solution on paper with handwriting; scan and merge into a single PDF. Rename your PDF as your student ID, for instance *26DS(E)-97.PDF*. Then upload.

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1. Sets and theri operations

Q1. Define a set. Give three examples of sets from real-life situations related to computer science.

Q2. Write the following sets in listed and set-builder form:

(a) $X = \{1, 3, 5, 7, 9\}$

(b) $Y = \{x \mid x \text{ is a letter in the word “PROGRAM”}\}$

Q3. If

$$A = \{x \mid x \text{ is a multiple of } 2,\ x \leq 10\},$$

$$B = \{x \mid x \text{ is a multiple of } 3,\ x \leq 10\},$$

find $A \cap B$, $A \cup B$, and $A - B$.

Q4. Construct sets A, B and C as follows: where a and b are first and second digit of our student id. For instance, your student id is 26DS(E)-97, then $a = 9$ and $b = 7$.

$$A = \{a,\ b,\ a+b,\ a-b,\ a+1,\ a-1\},$$

$$B = \{a,\ b,\ a+b+1,\ a-b+1,\ a+1,\ b+1\},$$

$$C = \{a+2,\ b+2,\ a+b+2,\ a-b+2,\ ab,\ a/b\},$$

Use set A, B and C in Questions 5 to 11.

Q5. find:

(a) $A \cup B$

(b) $A \cap B$

(c) $A - B$

(d) $B - A$

(e) $A \times B$

Q6. Draw Venn diagrams to illustrate:

(a) $(A \cup B) \cap C$

(b) $(A \cap B)'$

(c) $A - (B \cup C)$

Q7. Verify De Morgan's Laws for sets A and B.

Q8. Prove that set union and intersection are commutative and associative, whereas difference and Cartesian product are not commutative.

Note: You may prove this by taking any two sets out of the three sets A, B, and C defined in Q4.

Q9. How many subsets of sets A, B, and C (from Q3) are possible? Give only the number.

Q10. Find all subsets of the set $A \cap B \cap C$.

Q11. Determine the power set of $A \cap B \cap C$.

Q12. Give one real-world application of power sets in computer programming or databases.

Q13. List the identities of De Morgan's Laws and the distributive laws. Also prove them using a membership table.

2. Relations on Sets

Q14. Let $A = \{1, 2, 3\}$ and define $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$. Determine if R is reflexive, symmetric, and transitive.

Q15. Let $A = \{a, b, c\}$ and $B = \{1, 2\}$. List all possible relations from A to B.

Q16. Consider $R = \{(x, y) \mid x \text{ divides } y\}$ on $A = \{1, 2, 3, 4, 6\}$. List all elements of R . Also check if relation is equivalence?

Q17. Let R_1 and R_2 be relations on a set A. Prove or disprove each of the following.

(a) If R_1 and R_2 are reflexive, then $R_1 \cap R_2$ is reflexive.

(b) If R_1 and R_2 are symmetric, then $R_1 \cup R_2$ is symmetric.

(c) If R_1 and R_2 are transitive, then $R_1 \cup R_2$ is transitive.

(d) If R_1 is reflexive and R_2 is any relation on A, then $R_1 \circ R_2$ is reflexive.

(e) If R_1 and R_2 are equivalence relations, is $R_1 \cap R_2$ necessarily an equivalence relation?

Q18. Let $R = \{(a, b), (b, c), (a, c)\}$ on $A = \{a, b, c\}$. Is R transitive? Explain.

Q19. Define an equivalence relation. Give one example from real life.

Q20. If $R_1 = \{(1, 2), (2, 3)\}$ and $R_2 = \{(2, 1), (3, 2)\}$ on $\{1, 2, 3\}$, find $R_1 \circ R_2$ (composition of relations).

3. Functions and Their Properties

Q20. Define function, domain, co-domain, and range with examples.

Q21. Determine whether each of the following relations is a function:

(a) $\{(1, 2), (2, 4), (3, 9)\}$

(b) $\{(1, 2), (1, 3), (2, 4)\}$

Q22. If $f(x) = 2x + 3$, find $f(0)$, $f(2)$, and $f(-1)$.

Q23. Let $f(x) = x^2 + 1$ and $g(x) = x + 2$. Find:

(a) $(f \circ g)(x)$

(b) $(g \circ f)(x)$

Q24. Determine whether these functions are one-to-one, onto, or both:

(a) $f(x) = 3x - 2$

(b) $f(x) = x^2$

Q25. Let $f : A \to B$, where $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, and $f = \{(1, 4), (2, 5), (3, 6)\}$.

(a) Draw the mapping diagram.

(b) Find the inverse relation f^{-1}.

(c) Is f invertible? Why?

Q26. If $f(x) = 2x + 1$ and $g(x) = x - 3$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Q27. Find the inverse of each of the following functions:

(a) $f(x) = 3x + 5$

(b) $f(x) = \dfrac{x - 2}{4}$

(c) $f(x) = \dfrac{2x}{x + 1}$

Q28. Verify that $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ for $f(x) = 2x + 3$ and $g(x) = x - 1$.

End of Assignment